🟥 RS004 Week 3 — English Worksheet

Topic: Player & Keyboard Movement · **Course:** Platformer Game · **Time:** about 45 minutes

This week a **player** appears — a rectangle you control with the **arrow keys**. In the live lesson you wrote the code for it. On this worksheet you will think about and explain that code: how `pygame.Rect` tracks position, how `pygame.key.get_pressed()` reads held-down keys, and how the player is stopped from sliding off the screen.

> Keep these words handy: **Rect**, **position**, `get_pressed()`, **speed**, **boundary (edge)**.

1 · Predict 🔮

Read each snippet. Write what you think it does.

```
player = pygame.Rect(100, 500, 50, 50)
```

What do the four numbers mean? Where on the screen is the player?

```
keys = pygame.key.get_pressed()
if keys[pygame.K_RIGHT]:
    player.x = player.x + 5
```

The right arrow is held down. What happens to the player each frame?

```
if player.x > WIDTH - player.width:
    player.x = WIDTH - player.width
```

The player runs into the right wall. What does this code stop from happening?

2 · Rect Reading 📐

A player is `pygame.Rect(100, 500, 50, 50)`. Fill in the blanks.

```
player.x      = ___
player.y      = ___
player.width  = ___
player.right  = ___   (hint: x + width)
player.bottom = ___   (hint: y + height)
```

3 · Spot the Bug 🐛

Each snippet is broken. Write the fixed code, then explain why the original was wrong.

Bug A — Arrow keys should move the player smoothly while held. Right now the player only moves on a brand-new key press, one step at a time.

```
for event in pygame.event.get():
    if event.type == pygame.KEYDOWN:
        if event.key == pygame.K_RIGHT:
            player.x = player.x + 5
```

Hint: for *held* keys we use `pygame.key.get_pressed()`, not `KEYDOWN` events.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The player should stop at the left edge. Right now it disappears off the left side.

```
keys = pygame.key.get_pressed()
if keys[pygame.K_LEFT]:
    player.x = player.x - 5
```

Hint: after moving, check if `player.x` went below 0 and pull it back.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Both arrows should work. Right now only the right arrow does anything.

```
keys = pygame.key.get_pressed()
if keys[pygame.K_RIGHT]:
    player.x = player.x + 5
    if keys[pygame.K_LEFT]:
        player.x = player.x - 5
```

Hint: the left check is trapped *inside* the right check. They should be separate.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Explain the Code 📖

Here is a working player. Read it carefully and answer the questions below.

```
PLAYER_SPEED = 5
player = pygame.Rect(100, 500, 50, 50)

keys = pygame.key.get_pressed()

if keys[pygame.K_RIGHT]:
    player.x = player.x + PLAYER_SPEED
if keys[pygame.K_LEFT]:
    player.x = player.x - PLAYER_SPEED

if player.x < 0:
    player.x = 0
if player.x > WIDTH - player.width:
    player.x = WIDTH - player.width

pygame.draw.rect(screen, (255, 0, 0), player)
```

What does `PLAYER_SPEED` control, and what would happen if you made it bigger?

Why are the two arrow-key checks both `if` (not `if` / `elif`)? What does that let the player do?

The line `if player.x < 0: player.x = 0` — in your own words, what is it protecting against?

What does `WIDTH - player.width` mean, and why not just use `WIDTH`?

What is the last line doing, and what would the player look like without it?

5 · Explain Your Lesson Code 🎥

Explain the code **you** wrote in today's lesson. Record a short phone video — you may show it running while you talk. Try to use these words: **Rect**, **arrow keys**, **speed**, **edge**, **boundary**.

> 1. Show your player moving left and right.
> 2. Run into a wall and show it stops.
> 3. Read your boundary check out loud and explain what it does.
> 4. Point to where you set the speed and say what number you chose and why.

Write what you will say in your video. Plan it here before you record.

Submit ✅

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.